

EquiOrient: Supplementary Material

Anonymous Authors
Anonymous Institution

1. Full result matrices (Regime B, v2 corrected)

Validation accuracy at the selected lambda; held-out $V \circ H$ behavioral accuracy; accuracy with z zeroed (bias decode); paired both-correct; latent equivariance error on the held-out composition; depth probes (full- z and z_d -only); per-epoch answer/structural loss curves.

arm	val	holdout	z-corr	both	latent	probe	probe-zd
ordinary_sft	0.8875	—	—	—	—	—	—
augmentation	0.9917	1.0000	0.5000	1.0000	14.6516	0.5500	0.5167
output-consistency	0.9917	1.0000	0.5000	1.0000	19.2160	0.5500	0.5333
latent-invariance	0.9792	1.0000	0.5000	1.0000	5.5740	0.5500	0.5167
EquiOrient	0.9958	1.0000	0.5000	1.0000	0.0452	0.5333	0.5333
wrong-geometry	0.9917	1.0000	0.5000	1.0000	0.0261	0.5333	0.5500

Table 1. Regime B (5-object scenes; Modal L40S, 1357 s; repo commit 65c6c1c9). Source: results/equiorient/pilot_run_v2_final_matrix.json.

2. Full result matrices (Regime A, v1 corrected)

arm	latent ($V \circ H$)	probe	probe-zd
augmentation	10.3584	0.6111	—
EquiOrient	0.1717	0.5000	—
wrong-geometry	0.1572	0.5000	—

Table 2. Regime A (4-object scenes). Source: results/equiorient/pilot_run_v1_corrected_matrix.json.

3. Per-transform latent equivariance errors (correct vs. wrong rho)

Regime B, held-out scenes, mean $\|\rho(T)\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$:

arm	ρ_H	ρ_V	$\rho_{V \circ H}$	wrong-H	wrong-V	wrong- $V \circ H$
augmentation	14.8577	11.9853	14.6516	14.3586	10.3046	13.4313
output-consistency	17.0195	14.1305	19.2160	16.6655	12.6904	13.7582
latent-invariance	3.1233	3.4106	5.5740	3.4288	3.0393	0.8171
EquiOrient	0.0331	0.0062	0.0452	4.8945	5.0417	4.8316
wrong-geometry	3.1830	3.3059	0.0261	0.0198	0.0052	3.1691

Table 3. The composition metric cannot separate correct from wrong rho (the axis-swapped wrong laws compose to the correct action on $V \circ H$); per-transform errors can and do.

arm	ans e1	ans e2	struct e1	struct e2
augmentation	3.9746	0.2405	—	—
output-consistency	4.0411	0.1835	0.179039	0.161825
latent-invariance	4.0545	0.3734	0.166623	0.491841
EquiOrient	7.4582	1.2425	0.015562	0.013168
wrong-geometry	7.3594	1.2227	0.015286	0.014299

Table 4. The manipulation check: every structural arm has nonzero structural loss; EquiOrient and output-consistency decrease it, and the answer loss converges in all arms.

4. Training loss curves (per epoch, Regime B)

5. Depth probe: n , intervals, and the z_d block

Regime B: probe fit on validation-scene depth pairs (identity images), scored on hold-out-scene depth pairs, $n=60$ (3 scenes $\times$ 20 pairs); Wilson 95% CI for 0.5333 (32/60) is [0.41, 0.65]; chance 0.5. All arms are inside the interval. The z_d -block-only probe (0.52–0.55) is likewise flat, so the invariance-shaped depth component did not acquire decodable depth structure.

Regime A: $n=36$; EquiOrient 0.5000, augmentation 0.6111 (22/36, Wilson CI [0.45, 0.75]) – within noise of each other and of chance.

6. Algebra and data verification

- 21 unit tests on the transformation algebra (identity, inverse, composition, no-contradiction, determinism, wrong-geometry reachability). - 10,880 law checks on the v2 scene pack (17 scenes $\times$ 4 transforms $\times$ 20 ordered pairs $\times$ 8 relations), 0 violations; v1 pack: 4,896 rows, 0 violations; hostile renderer-bug test detects injected violations. - Human spot check of 51 synthetic pairs: 0 flags (recorded verdict).

7. Compute ledger and provenance

run	platform	outcome	GPU ti
v1 pilot (4 runs)	A6000 (Thunder)	voided: implementation deviations (logged)	2.7 h
Amendment-D pilot	L40S (Modal)	voided: identically-zero structural losses (manipulation-check failure, decision log)	9 min
corrected v2	L40S (Modal)	VALID	1357 s
corrected v1	L40S (Modal)	VALID	910 s

Table 5. Voided runs are excluded from all scientific tables; the failure class (zero structural loss) is asserted against in the released harness.

8. Reproducibility

- Harness: scripts/equiorient/pilot_harness.py (single file, no analyzer imports; CPU tiny gate runs the full pipeline end-to-end including the manipulation check before any GPU spend). - Generators: src/equiorient/datasets.py (v1/v2 recipes), scripts/equiorient/build_pilot_data.py (deterministic seeds 20260814/20260815). - Frozen protocol: configs/equiorient_pilot_freeze.yaml and _v2.yaml; decision log: research/EQUIORIENT_DECISION_LOG.md. - Execution pipeline: modal/equiorient_modal.py (serverless, pinned stack torch 2.8.0+cu128 / transformers==4.57.6 / Qwen3-VL-8B @ 0c351dd0); full pilot <25 min GPU.